

03/09/25

UNIT-I

MATHEMATICAL LOGIC

* PROPOSITIONS:

Let us consider the following statements -

- (i) Bangalore is in Karnataka
- (ii) 7 is a prime number
- (iii) 5 is divisible by 3
- (iv) Every rectangle is a square.

→ Each of the above sentences is a statement which can be either true/false, but not both.

→ Such sentences are called "Propositions".

→ Proposition is a statement [declaration], which in a given context can be said to be either true or false, but not both.

→ It should be noted that all sentences are not propositions.

for eg: (i) $xy = yx$

(ii) Read this carefully

→ Propositions are usually represented by small letters such as p, q, r, s .

→ The truth or falsity of proposition is called its truth value.

→ If a proposition is true, we indicate its truth value by the symbol 1 (T), and if it is false, by the symbol 0 (F).

* LOGICAL CONNECTIVES & TRUTH TABLES

→ The words like 'and', 'or', 'not', 'if-then', 'if & only if' which are used to form a new proposition by connecting two or more propositions is called compound proposition and the words are called logical connectives.

→ The original proposition from which the compound proposition are obtained are called components/ primitives.

→ Propositions which do not contain any logical connective are called simple proposition.

1) Negation:

• A proposition obtained by using the word 'not' in appropriate place is called the negation of the given proposition.

• The negation of proposition p is denoted by $\sim p$.

• Truth table is given by -

p	$\sim p$
0	1
1	0

2) Conjunction:

• The compound proposition obtained by combining 2 propositions p, q using the word 'and' b/w them is called conjunction.

• Denoted by - $p \wedge q$

• Truth table -

p	q	$p \wedge q$
0	0	0
0	1	0
1	0	0
1	1	1

3) Disjunction:

• The compound proposition obtained by combining 2 propositions p, q using the word 'or' b/w them is called disjunction.

• Denoted by - $p \vee q$

• Truth table -

p	q	$p \vee q$
0	0	0
0	1	1
1	0	1
1	1	1

4) Conditional:

• Compound Proposition obtained by combining 2 propositions p, q using the word 'if-then' at appropriate place is called conditional.

• Denoted by - $p \rightarrow q$

• Truth table -

p	q	$p \rightarrow q$
0	0	1
0	1	1
1	0	0
1	1	1

5) Bi-Conditional:

• Compound Proposition is obtained by combining 2 given propositions p, q using the word 'if & only if' at appropriate place is called bi-conditional.

• Denoted by - $p \leftrightarrow q$

• Truth table -

p	q	$p \leftrightarrow q$
0	0	1
0	1	0
1	0	0
1	1	1

Q) Construct the truth tables for the following compound propositions -

(i) $p \wedge (\neg q)$

(ii) $(\neg p) \vee q$

(iii) $p \rightarrow (\neg q)$

Solⁿ:

(i) $p \wedge (\neg q)$

p	q	$\neg q$	$p \wedge (\neg q)$
0	0	1	0
0	1	0	0
1	0	1	1
1	1	0	0

(ii) $(\neg p) \vee q$

p	q	$\neg p$	$(\neg p) \vee q$
0	0	1	1
0	1	1	1
1	0	0	0
1	1	0	1

(iii) $p \rightarrow (\neg q)$

p	q	$\neg q$	$p \rightarrow (\neg q)$
0	0	1	1
0	1	0	1
1	0	1	1
1	1	0	0

(iv) $p \wedge (q \wedge p)$

p	q	$q \wedge p$	$p \wedge (q \wedge p)$
0	0	0	0
0	1	0	0
1	0	0	0
1	1	1	1

(v) $p \wedge (q \vee p)$

p	q	$q \vee p$	$p \wedge (q \vee p)$
0	0	0	0
0	1	1	0
1	0	1	1
1	1	1	1

(vi) $(p \wedge q) \rightarrow \sim r$

p	q	r	$p \wedge q$	$\sim r$	$(p \wedge q) \rightarrow \sim r$
0	0	0	0	1	1
0	0	1	0	0	1
0	1	0	0	1	1
0	1	1	0	0	1
1	0	0	0	1	1
1	0	1	0	0	1
1	1	0	1	1	1
1	1	1	1	0	0

(vii) $q \wedge [(\sim r) \rightarrow p]$

p	q	r	$\sim r$	$\sim r \rightarrow p$	$q \wedge [(\sim r) \rightarrow p]$
0	0	0	1	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	0	1	0
1	1	0	1	1	1
1	1	1	0	1	1

(viii) $q \leftrightarrow [(\sim p \vee \sim q)]$

p	q	$\sim p$	$\sim q$	$\sim p \vee \sim q$	$q \leftrightarrow [(\sim p \vee \sim q)]$
0	0	1	1	1	0
0	1	1	0	1	1
1	0	0	1	1	0
1	1	0	0	0	0

$$(ix) [q \wedge (p \rightarrow r)] \rightarrow p$$

p	q	r	$p \rightarrow r$	$q \wedge (p \rightarrow r)$	$[q \wedge (p \rightarrow r)] \rightarrow p$
0	0	0	1	0	1
0	0	1	1	0	1
0	1	0	1	1	0
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	1
1	1	0	0	0	1
1	1	1	1	1	1

Q2) Express the following compound propositions in words.

$\Rightarrow p$: A circle is a conic

q : $\sqrt{5}$ is a real number

r : Exponential series is convergent.

(i) $p \wedge \sim q$

(ii) $\sim p \vee q$

(iii) $q \rightarrow \sim p$

(iv) $\sim p \leftrightarrow r$

Solⁿ (i) A circle is a conic and $\sqrt{5}$ is not a real number.

(ii) A circle is not a conic or $\sqrt{5}$ is a real number.

(iii) ~~If~~ $\sqrt{5}$ is a real number then a circle is not a conic.

(iv) A circle is not a conic iff and only if exponential series is convergent.

$\Rightarrow$ p : ΔABC is isosceles
 q : ΔABC is equilateral
 r : ΔABC is equiangular

(i) $q \rightarrow p$

If ΔABC is equilateral then ΔABC is isosceles.

(ii) $\sim p \rightarrow \sim q$

If ΔABC is not isosceles then ΔABC is not equilateral.

(iii) $q \rightarrow r$

If ΔABC is equilateral then ΔABC is equiangular.

(iv) $p \wedge \sim q$

ΔABC is isosceles and ΔABC is not equilateral.

(v) $\sim p \wedge q$

ΔABC is not isosceles and ΔABC is equilateral.

(vi) $p \leftrightarrow r$

ΔABC is isosceles iff ΔABC is equiangular.

Q3) Let p & q be primitive statements for which $p \rightarrow q$ is false. Determine the truth values for the following-

(i) $p \wedge q$ (ii) $\sim p \vee q$ (iii) $q \rightarrow p$ (iv) $(\sim q \rightarrow \sim p)$

Solⁿ - By data, $p \rightarrow q$ is false, i.e. p is true & q is false.
 $\therefore$ The truth table is as follows -

p	q	$\sim p$	$\sim q$	$p \wedge q$	$\sim p \vee q$	$q \rightarrow p$	$\sim q \rightarrow \sim p$
1	0	0	1	0	0	1	0

Q4) Given that, p, q, r are propositions having truth values 0, 0, 1 respectively. Find the truth values of the following compound propositions.

- Solⁿ
- $p \rightarrow (q \wedge r)$
 - $(p \vee q) \vee r$
 - $(p \wedge q) \rightarrow r$
 - $p \rightarrow [q \rightarrow (\sim r)]$

Solⁿ - By data,
 p is 0
 q is 0
 r is 1

(i)

p	q	r	$q \wedge r$	$p \rightarrow (q \wedge r)$
0	0	1	0	1

(ii)

p	q	r	$p \vee q$	$(p \vee q) \vee r$
0	0	1	0	1

(iii)

p	q	r	$p \wedge q$	$(p \wedge q) \rightarrow r$
0	0	1	0	1

(iv)

p	q	r	$\sim r$	$q \rightarrow (\sim r)$	$p \rightarrow [q \rightarrow (\sim r)]$
0	0	1	0	1	1

*TAUTOLOGY & CONTRADICTION:

- $\rightarrow$ A compound proposition which is always true for all possible truth values of its components is called 'Tautology'.
 - $\rightarrow$ A compound proposition which is always false for all possible truth values of its components is called 'Contradiction'.
 - $\rightarrow$ A compound proposition which is neither tautology nor contradiction is called 'Contingency'.
- Q1) By construction truth table, show that for any propositions p, q, r , the compound proposition $[C(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is a tautology.

Solⁿ Truth Values

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r)$	$p \rightarrow r$	$[C(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$
0	0	0	1	1	1	1	1
0	0	1	1	1	1	1	1
0	1	0	0	0	0	1	1
0	1	1	1	1	1	1	1
1	0	0	0	1	0	0	1
1	0	1	0	1	0	0	1
1	1	0	1	0	0	0	1
1	1	1	1	1	1	1	1

Hence, the compound proposition is a tautology.

Q2) Show that the compound proposition -
 $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$ is a tautology.

Soln -

p	q	r	$p \vee q$	$p \rightarrow r$	$q \rightarrow r$	$(p \vee q) \wedge (p \rightarrow r)$	$A \wedge B$
0	0	0	0	1	1	0	0
0	0	1	0	1	1	0	0
0	1	0	1	1	0	1	1
0	1	1	1	1	1	1	1
1	0	0	1	0	1	0	0
1	0	1	1	1	1	1	1
1	1	0	1	0	0	0	0
1	1	1	1	1	1	1	1

$B \rightarrow r$
1
1
1
1
1
1
1
1

$\therefore$ It is a tautology.

Q3) By constructing the truth table, examine whether the compound proposition $[p \rightarrow (q \rightarrow r)] \leftrightarrow [(p \rightarrow q) \wedge (q \rightarrow r)]$ is tautology/contradiction.

p	q	r	$q \rightarrow r$	$p \rightarrow (q \rightarrow r)$	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow q \wedge q \rightarrow r$
0	0	0	1	1	1	1	1
0	0	1	1	1	1	1	1
0	1	0	0	1	0	0	0
0	1	1	1	1	1	1	1
1	0	0	1	0	0	0	0
1	0	1	1	0	0	0	0
1	1	0	0	0	1	0	0
1	1	1	1	1	1	1	1

$A \leftrightarrow B$
1
1
0
1
0
0
0
1

$\therefore$ It is not a tautology/contradiction.

Q4) Examine whether $[p \vee q] \rightarrow r \leftrightarrow [r \rightarrow (p \vee q)]$ is a tautology or not.



P	q	r	$p \vee q$	$(p \vee q) \rightarrow r$	$\sim r$	$\sim(p \vee q)$	$\sim r \rightarrow \sim(p \vee q)$
0	0	0	0	1	1	1	1
0	0	1	0	0	0	1	0
0	1	0	1	0	1	0	1
0	1	1	1	1	0	0	0
1	0	0	1	0	1	0	1
1	0	1	1	1	0	0	0
1	1	0	1	0	1	0	1
1	1	1	1	1	0	0	0

$A \leftrightarrow B$
1
1
1
1
1
1
1
1

$\therefore$ It is a tautology.

Q5) Show that the compound proposition is a tautology by using truth table.

(i) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$

(ii) $[(p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow (p \vee q) \rightarrow r$

Q6) For any propositions p, q, show that the compound proposition is a contradiction.

(i) $p \wedge (\sim p \wedge q)$

(ii) $(p \vee q) \wedge (\sim p) \wedge (\sim q)$

Sol (i)

p	q	$\sim p$	$\sim p \wedge q$	$p \wedge (\sim p \wedge q)$
0	0	1	0	0
0	1	1	1	0
1	0	0	0	0
1	1	0	0	0

$\therefore$ It is a contradiction.

Sol (ii)

p	q	$p \vee q$	$\sim p$	$\sim q$	$(p \vee q) \wedge (\sim p) \wedge (\sim q)$
0	0	0	1	1	0
0	1	1	1	0	0
1	0	1	0	1	0
1	1	1	0	0	0

$\therefore$ It is a contradiction.



HN
sol 5)

(i) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r)$	$p \rightarrow r$	$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$
0	0	0	1	1	1	1	1
0	0	1	1	1	1	1	1
0	1	0	0	1	0	1	1
0	1	1	1	1	1	1	1
1	0	0	0	1	0	0	1
1	0	1	0	1	0	0	1
1	1	0	1	0	0	1	1
1	1	1	1	1	1	1	1

$\therefore$ It is a tautology.

(ii) $[(p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow [(p \vee q) \rightarrow r]$

p	q	r	$p \rightarrow r$	$q \rightarrow r$	$(p \rightarrow r) \wedge (q \rightarrow r)$	$p \vee q$	$(p \vee q) \rightarrow r$	$[(p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow [(p \vee q) \rightarrow r]$
0	0	0	1	1	1	0	0	0
0	0	1	1	1	1	0	1	1
0	1	0	1	0	0	1	0	0
0	1	1	1	1	1	1	1	1
1	0	0	0	1	0	1	0	0
1	0	1	1	1	1	1	1	1
1	1	0	0	0	0	1	0	0
1	1	1	1	1	1	1	1	1

$(p \vee q) \rightarrow r$	$x \rightarrow y$
1	1
0	1
1	1
0	1
1	1
0	1
1	1
1	1

$\therefore$ It is a tautology.

11/07/25

* LOGICAL EQUIVALENCE:

$\rightarrow$ A compound propositions U & V are said to be logically equivalent if U & V have the same truth values for all possible truth values of its components.

$\rightarrow$ It is denoted by $U \Leftrightarrow V$

$\rightarrow$ In other words, U & V are said to be logically equivalent if $U \Leftrightarrow V$ is a tautology.

Q1) For any 2 propositions, prove the following -

(i) $p \rightarrow q \Leftrightarrow \sim p \vee q$

(ii) $p \rightarrow \sim q \Leftrightarrow q \rightarrow \sim p$

(iii) $\sim(p \leftrightarrow q) \Leftrightarrow \{(p \vee q) \wedge [\sim(p \wedge q)]\}$

Sol (i) - Truth table -

p	q	$p \rightarrow q$	$\sim p$	$\sim p \vee q$	$(p \rightarrow q) \Leftrightarrow \sim p \vee q$
0	0	1	1	1	1
0	1	1	1	1	1
1	0	0	0	0	1
1	1	1	0	1	1

We note that, $p \rightarrow q$ & $\sim p \vee q$ are identical.

Hence, $p \rightarrow q \Leftrightarrow \sim p \vee q$.

9

Solⁿ (iii)

p	q	$\sim q$	$p \rightarrow \sim q$	$\sim p$	$q \rightarrow \sim p$	$p \rightarrow \sim q \leftrightarrow q \rightarrow \sim p$
0	0	1	1	1	1	1
0	1	0	1	0	0	0
1	0	1	0	1	1	1
1	1	0	0	0	0	0

Hence, $p \rightarrow \sim q$ & $q \rightarrow \sim p$ are identical.
 $\therefore p \rightarrow \sim q \leftrightarrow q \rightarrow \sim p$

Solⁿ (ii)

p	q	$p \leftrightarrow q$	$\sim(p \leftrightarrow q)$	$p \vee q$	$p \wedge q$	$\sim(p \wedge q)$
0	0	1	0	0	0	1
0	1	0	1	1	0	1
1	0	0	1	1	0	1
1	1	1	0	1	1	0

$[(p \vee q) \wedge \sim(p \wedge q)]$	$\sim(p \leftrightarrow q) \leftrightarrow [(p \vee q) \wedge \sim(p \wedge q)]$
0	1
1	1
1	1
0	1

Hence, $\sim(p \leftrightarrow q)$ & $[(p \vee q) \wedge \sim(p \wedge q)]$ are identical
 $\therefore \sim(p \leftrightarrow q) \leftrightarrow [(p \vee q) \wedge \sim(p \wedge q)] //$

Q2) By constructing the TT, show that -
 $[(p \vee q) \rightarrow r] \leftrightarrow (p \rightarrow r) \wedge (q \rightarrow r)$

Solⁿ -

p	q	r	$p \vee q$	$p \vee q \rightarrow r$	$p \rightarrow r$	$q \rightarrow r$	$p \rightarrow r \wedge q \rightarrow r$
0	0	0	0	1	1	1	1
0	0	1	0	1	1	1	1
0	1	0	1	0	1	0	0
0	1	1	1	1	1	1	1
1	0	0	1	0	0	1	0
1	0	1	1	1	1	1	1
1	1	0	1	0	0	0	0
1	1	1	1	1	1	1	1

Hence, $(p \vee q) \rightarrow r$ & $(p \rightarrow r) \wedge (q \rightarrow r)$ are identical
 $\therefore [(p \vee q) \rightarrow r] \leftrightarrow (p \rightarrow r) \wedge (q \rightarrow r)$

Q3) By constructing truth table, ST -

- $[(p \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge (r \leftrightarrow p)] \leftrightarrow [(p \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow p)]$
- $[p \rightarrow (q \rightarrow r)] \leftrightarrow (p \wedge q) \rightarrow r$
- $[\sim p \wedge (\sim q \wedge r)] \vee [(q \wedge r) \vee (\sim q \wedge \sim r)] \leftrightarrow r$

10

Sol (i)

p	q	r	$p \leftrightarrow q$	$q \leftrightarrow r$	$r \leftrightarrow p$	$(p \leftrightarrow q) \wedge (q \leftrightarrow r)$
0	0	0	1	1	1	1
0	0	1	1	0	0	0
0	1	0	0	0	1	0
0	1	1	0	1	0	0
1	0	0	0	1	0	0
1	0	1	0	0	1	0
1	1	0	1	0	0	0
1	1	1	1	1	1	1

$(p \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge (r \leftrightarrow p)$	$p \rightarrow q$	$q \rightarrow r$	$r \rightarrow p$
---	-------------------	-------------------	-------------------

1	1	1	1
0	1	0	0
0	1	1	0
0	0	1	1
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1

$(p \rightarrow q) \wedge (q \rightarrow r)$	$(p \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow p)$	$(p \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge (r \leftrightarrow p)$
--	---	---

1	1	1
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0

LHS=RHS

Hence, $[(p \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge (r \leftrightarrow p)] \leftrightarrow [(p \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow p)]$

Sol (ii)

p	q	r	$q \rightarrow r$	LHS $p \rightarrow (q \rightarrow r)$	RHS $(p \wedge q) \rightarrow r$
0	0	0	1	1	1
0	0	1	1	1	1
0	1	0	0	1	1
0	1	1	1	1	1
1	0	0	1	1	0
1	0	1	1	1	0
1	1	0	0	0	0
1	1	1	1	1	1

$\therefore$ LHS=RHS

Hence, $p \rightarrow (q \rightarrow r) \leftrightarrow (p \wedge q) \rightarrow r$

Sol (iii)

p	q	r	$\neg q$	$\neg q \wedge r$	$\neg p$	$\neg p \wedge (\neg q \wedge r)$	$q \wedge r$	$p \wedge r$
0	0	0	1	0	1	0	0	0
0	0	1	1	1	1	1	0	0
0	1	0	0	0	1	0	0	0
0	1	1	0	0	1	0	1	0
1	0	0	1	0	0	0	0	0
1	0	1	1	1	0	1	0	0
1	1	0	0	0	0	0	0	0
1	1	1	0	0	0	0	1	1

$(q \vee r) \vee (p \wedge r)$	LHS $\neg p \vee (q \wedge r)$
--------------------------------	--------------------------------

1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1

$\therefore$ LHS=RHS

Hence, $[(\neg p \wedge (\neg q \wedge r)) \vee (q \vee r) \vee (p \wedge r)] \leftrightarrow r$

* CONVERSE, INVERSE, CONTRAPOSITIVE:

- Consider the conditional statement $p \rightarrow q$, then
- (i) $q \rightarrow p$ is called the converse of $p \rightarrow q$.
 - (ii) $\neg p \rightarrow \neg q$ is called inverse of $p \rightarrow q$.
 - (iii) $\neg q \rightarrow \neg p$ is called contrapositive of $p \rightarrow q$.

Truth table:

p	q	$p \rightarrow q$	$q \rightarrow p$	$\neg p$	$\neg q$	$\neg p \rightarrow \neg q$	$\neg q \rightarrow \neg p$
0	0	1	1	1	1	1	1
0	1	1	0	1	0	0	1
1	0	0	1	0	1	1	0
1	1	1	1	0	0	1	1

From the table we note that,

$$p \rightarrow q \iff \neg q \rightarrow \neg p \text{ \& }$$

$$q \rightarrow p \iff \neg p \rightarrow \neg q$$

- Q) Define the converse, inverse, contrapositive of $p \rightarrow q$ with truth table. Hence, write the converse, inverse & contrapositive of the following statement.
- If Kabir wears brown pant then he will wear white shirt.

Solⁿ By data,

p : Kabir wears brown pant.

q : He will wear white shirt.

- (i) Converse: $q \rightarrow p$ - If Kabir wears white shirt then he will wear brown pant.

- (ii) Inverse: $\neg p \rightarrow \neg q$ - If Kabir does not wear brown pant then he will not wear white shirt.

- (iii) Contrapositive: $\neg q \rightarrow \neg p$ - If Kabir does not wear white shirt then he will not wear brown pant.

- Q) Write the converse, inverse & contrapositive of the following statement -
- If a triangle is not isosceles then it is not equilateral.

Solⁿ p : A triangle is not isosceles.

q : A triangle is not equilateral.

- (i) Converse: $q \rightarrow p$ - If a triangle is not equilateral then it is not isosceles.

- (ii) Inverse: $\neg p \rightarrow \neg q$ - If a triangle is isosceles then it is equilateral.

- (iii) Contrapositive: $\neg q \rightarrow \neg p$ - If a triangle is equilateral then it is isosceles.

- Q) If a real no. x^2 is greater than zero, then x is not equal to zero.

Solⁿ p : A real no. x^2 is greater than zero

q : x is not equal to zero

- (i) Converse: $q \rightarrow p$ - If x is not equal to zero then x^2 is greater than zero.

- (ii) Inverse: $\neg p \rightarrow \neg q$ - If a real no. x^2 is not greater than zero, then x is equal to zero.

- (iii) Contrapositive: $\neg q \rightarrow \neg p$ - If a real no. x is equal to zero then x^2 is not greater than zero.

8) Write the converse, inverse & contrapositive -
 i) If ΔABC is an isosceles triangle then it has 2 equal sides.

ii) If a quadrilateral is a parallelogram then the diagonals bisect each other.

* LAWS OF LOGIC:

Laws of logic follows the definition of logical equivalence.

① Laws of double negation.
 $\sim(\sim p) \Leftrightarrow p$

② Commutative law

$$p \wedge q \Leftrightarrow q \wedge p$$

$$p \vee q \Leftrightarrow q \vee p$$

③ Associative law

$$p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$$

$$p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r$$

④ Distributive Law

$$p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$$

$$p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$$

⑤ Identity law

$$p \vee F_0 \Leftrightarrow p$$

$$p \wedge T_0 \Leftrightarrow p$$

Tautology

⑥ Inverse law

$$p \vee (\sim p) \Leftrightarrow T_0$$

$$p \wedge (\sim p) \Leftrightarrow F_0$$

⑦ Domination law

$$p \vee T_0 \Leftrightarrow T_0$$

$$p \wedge F_0 \Leftrightarrow F_0$$

⑧ Idempotent law

$$p \wedge p \Leftrightarrow p$$

$$p \vee p \Leftrightarrow p$$

⑨ Absorption law

$$p \vee (p \wedge q) \Leftrightarrow p$$

$$p \wedge (p \vee q) \Leftrightarrow p$$

⑩ De-morgan's law

$$\sim(p \vee q) \Leftrightarrow \sim p \wedge \sim q$$

$$\sim(p \wedge q) \Leftrightarrow \sim p \vee \sim q$$

where, $T_0 \rightarrow$ tautology

$F_0 \rightarrow$ contradiction

Note: $p \rightarrow q \Leftrightarrow \sim p \vee q$

2

HW

Q) p : ΔABC is an isosceles triangle
 q : ΔABC has 2 equal sides.

(i) Converse: $q \rightarrow p$ - If ΔABC has 2 equal sides then it is an isosceles triangle.

(ii) Inverse: $\neg p \rightarrow \neg q$ - If ΔABC is not an isosceles triangle then it does not have 2 equal sides.

(iii) Contrapositive: $\neg q \rightarrow \neg p$ - If ΔABC does not have 2 equal sides then ~~it is~~ it is not an isosceles triangle.

Q) p : A quadrilateral is a parallelogram
 q : Its diagonals bisect each other.

(i) Converse: $q \rightarrow p$ - If a quadrilateral's diagonals bisect each other then it is a parallelogram.

(ii) Inverse: $\neg p \rightarrow \neg q$ - If a quadrilateral is not a parallelogram then its diagonals do not bisect each other.

(iii) Contrapositive: $\neg q \rightarrow \neg p$ - If a quadrilateral's diagonals do not bisect each other then it is not a parallelogram.

17/09/25

Q) Prove the following logical equivalence using laws of logic -

$$p \vee [p \wedge (p \vee q)] \leftrightarrow p$$

Sol - Consider LHS:

$$p \vee [p \wedge (p \vee q)]$$

$$\Leftrightarrow p \vee p \quad \therefore \text{by absorption law}$$

$$\Leftrightarrow p \quad \therefore \text{by idempotent law}$$

$$= \text{RHS} //$$

$$p \vee [p \vee q] \wedge (p \vee \neg q) \vee q \Leftrightarrow p \vee q$$

Sol - Consider LHS:

$$[p \vee q] \wedge (p \vee \neg q) \vee q$$

$$\Leftrightarrow [p \vee (q \wedge \neg q)] \vee q \quad \therefore \text{by distributive law}$$

$$\Leftrightarrow [p \vee F_0] \vee q, \text{ by inverse law}$$

$$\Leftrightarrow p \vee q, \text{ by identity law.}$$

$$= \text{RHS} //$$

$$3) [p \rightarrow (q \rightarrow r)] \Leftrightarrow (p \wedge q) \rightarrow r$$

Sol - Consider LHS:

$$p \rightarrow (q \rightarrow r)$$

$$\Leftrightarrow p \rightarrow (\neg q \vee r) \quad \text{using } p \rightarrow q \Leftrightarrow \neg p \vee q$$

$$\Leftrightarrow \neg p \vee (\neg q \vee r)$$

$$\Leftrightarrow (\neg p \vee \neg q) \vee r, \text{ using Associative law}$$

$$\Leftrightarrow \neg(p \wedge q) \vee r, \text{ De-Morgan's law.}$$

$$\Leftrightarrow (p \wedge q) \rightarrow r, \text{ using } p \rightarrow q \Leftrightarrow \sim p \vee q \\ = \text{RHS}$$

$$4) [(p \rightarrow q) \wedge (p \rightarrow \sim q)] \Leftrightarrow \sim p$$

Solⁿ - Consider LHS:

$$\Leftrightarrow [(\sim p \vee q) \wedge (\sim p \vee \sim q)], \text{ using } p \rightarrow q \Leftrightarrow \sim p \vee q$$

$$= [(\sim p \vee q) \wedge (\sim p \vee \sim q)], \text{ Using De-Morgan's law}$$

$$\Leftrightarrow \sim p \vee (q \wedge \sim q), \text{ Using Distributive law}$$

$$\Leftrightarrow \sim p \vee F, \text{ Using Inverse law}$$

$$\Leftrightarrow \sim p, \text{ Using Identity law}$$

$$= \text{RHS}$$

$$5) [(\sim p \vee \sim q) \rightarrow (p \wedge q \wedge r)] \Leftrightarrow p \wedge q$$

consider LHS:

$$\text{Solⁿ } \Leftrightarrow [(\sim p \vee \sim q) \rightarrow (p \wedge q \wedge r)]$$

$$\Leftrightarrow \sim(\sim p \vee \sim q) \vee (p \wedge q \wedge r), \text{ using } p \rightarrow r \Leftrightarrow \sim p \vee r$$

$$\Leftrightarrow \sim \sim (p \wedge q) \vee (p \wedge q \wedge r), \text{ De-Morgan's law}$$

$$\Leftrightarrow (p \wedge q) \vee (p \wedge q \wedge r), \text{ Double negation}$$

$$\Leftrightarrow (p \wedge q) \vee ((p \wedge q) \wedge r), \text{ By associative law}$$

$$\Leftrightarrow p \wedge q, \text{ By absorption law}$$

$$= \text{RHS}$$

$$6) (p \rightarrow q) \wedge [\sim q \wedge (r \vee \sim q)] \Leftrightarrow \sim (q \vee p)$$

Solⁿ - LHS:

$$(p \rightarrow q) \wedge [\sim q \wedge (r \vee \sim q)]$$

$$\Leftrightarrow (\sim p \vee q) \wedge [\sim q \wedge (r \vee \sim q)], \text{ by using } p \rightarrow q \Leftrightarrow \sim p \vee q$$

$$\Leftrightarrow (\sim p \vee q) \wedge [\sim q \wedge (\sim q \vee r)], \text{ Commutative law}$$

$$\Leftrightarrow (\sim p \vee q) \wedge \sim q, \text{ By Absorption law}$$

$$\Leftrightarrow (\sim p \wedge \sim q) \vee (q \wedge \sim q), \text{ By distributive law}$$

$$\Leftrightarrow (\sim p \wedge \sim q) \vee F, \text{ By Inverse law}$$

$$\Leftrightarrow \sim p \wedge \sim q, \text{ By identity law}$$

$$\Leftrightarrow \sim (p \vee q), \text{ By De-Morgan's law}$$

$$\Leftrightarrow \sim (q \vee p), \text{ By commutative law}$$

$$7) [\sim p \wedge (\sim q \wedge r)] \vee (q \wedge r) \vee (p \wedge r) \Leftrightarrow r$$

Solⁿ - LHS:

$$[\sim p \wedge (\sim q \wedge r)] \vee (q \wedge r) \vee (p \wedge r)$$

$$= [(\sim p \wedge \sim q) \wedge r] \vee (q \wedge r) \vee (p \wedge r)$$

$$\Leftrightarrow [(\sim p \wedge \sim q) \wedge r] \vee [(q \wedge r) \vee (p \wedge r)], \text{ by Associative law}$$

$$\Leftrightarrow [\sim (p \vee q) \wedge r] \vee [(q \vee p) \wedge r], \text{ by De-Morgan's law \& Distributive law}$$

$$\Leftrightarrow [\sim p \vee q] \vee [q \vee p] \wedge r, \text{ by Distributive law}$$

$$\Leftrightarrow (\sim p \vee q) \vee (p \vee q) \wedge r, \text{ by Commutative law}$$

$\Leftrightarrow To \wedge r$, by inverse law.
 $\Leftrightarrow r$, by identity law
 $= R.H.S.$

18/09/20

* APPLICATIONS TO SWITCHING NETWORK-

→ A switching network is made up of wires and switches connecting 2 terminals A & B.

→ In such a network, if a switch is open [no current flow] or if the switch is closed [current flow].

→ If the switch P is open, we assign the symbol 0 & if the switch P is closed, we assign the symbol 1 to it.

→ Fig 1 shows a network having only one switch P. The current flows through the terminal A to B if it is closed.

→ This network is represented by P.

Fig-1: A — P — B

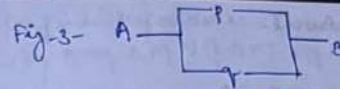
→ Fig 2 shows a series network consisting of 2 switches P & Q in which the current flows from A to B if both P & Q are closed.

→ This network is represented by $P \wedge Q$.

Fig-2: A — P — Q — B

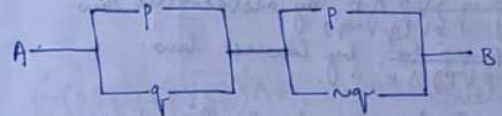
→ Fig 3 shows a parallel network consisting of 2 switches P, Q in which the current flows from A to B if $P \vee Q$ or both are closed.

→ This network is represented by $P \vee Q$.



→ By using the concept of laws of logic we can obtain a new network which is equivalent to given network and it contains less no. of switches.

Q) Simplify the following switching network using laws of logic.



Solⁿ - Compound propositions -

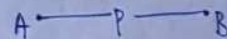
$$(P \vee Q) \wedge (P \vee \neg Q)$$

$$\Leftrightarrow P \vee (Q \wedge \neg Q), \text{ by distributive law.}$$

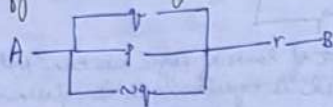
$$\Leftrightarrow P \vee F_0, \text{ by inverse law}$$

$$\Leftrightarrow P, \text{ by identity law.}$$

∴ The equivalent network is -



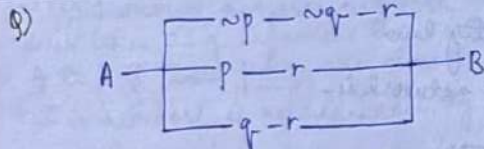
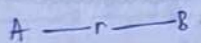
9) Simplify the switching network-



Solⁿ Compound Proposition-

$$\begin{aligned}
 & (q \vee p \vee \sim q) \wedge r \\
 \Leftrightarrow & (p \vee q \vee \sim q) \wedge r, \text{ commutative law} \\
 \Leftrightarrow & (p \vee (q \vee \sim q)) \wedge r, \text{ by associative law} \\
 \Leftrightarrow & (p \vee 1) \wedge r, \text{ by inverse law} \\
 \Leftrightarrow & 1 \wedge r, \text{ by domination law} \\
 \Leftrightarrow & r, \text{ by identity law}
 \end{aligned}$$

$\therefore$ The equivalent network is -



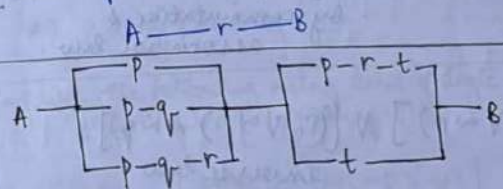
Simplify using laws of logic.

IT

Solⁿ Compound Propositions:

$$\begin{aligned}
 & (\sim p \wedge \sim q \wedge r) \vee (p \wedge r) \vee (q \wedge r) \\
 \Leftrightarrow & r \vee (\sim p \wedge \sim q \wedge r) \vee (p \wedge r) \vee (q \wedge r) \\
 \Leftrightarrow & [(\sim p \wedge \sim q) \wedge r] \vee [(p \wedge r) \vee (q \wedge r)], \text{ by Associative law} \\
 \Leftrightarrow & [\sim(p \vee q) \wedge r] \vee [(p \vee q) \wedge r], \text{ by De Morgan's law} \\
 \Leftrightarrow & (\sim(p \vee q) \vee (p \vee q)) \wedge r, \text{ by distributive law} \\
 \Leftrightarrow & 1 \wedge r, \text{ by inverse law} \\
 \Leftrightarrow & r, \text{ identity law}
 \end{aligned}$$

$\therefore$ The equivalent network is -

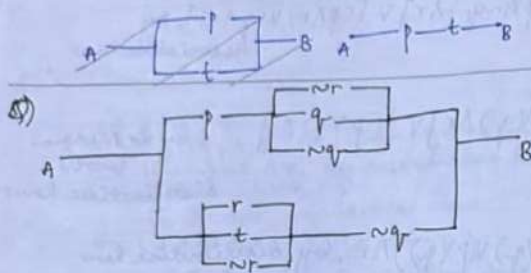


9)

$$\begin{aligned}
 \text{Solⁿ CP: } & [p \vee (p \wedge q) \vee (p \wedge q \wedge r)] \wedge [(p \wedge r \wedge t) \vee t] \\
 \Leftrightarrow & [(p \vee (p \wedge q)) \vee (p \wedge q \wedge r)] \wedge [(p \wedge r \wedge t) \vee t], \text{ by associative law} \\
 \Leftrightarrow & [p \vee (p \wedge q \wedge r)] \wedge [t], \text{ Absorption law, Associative law}
 \end{aligned}$$

$\Rightarrow P \wedge t$, absorption law.

$\therefore$ The equivalent network is:



Solⁿ - CP: $[p \wedge (\sim r \vee q \vee \sim q)] \vee [(r \vee t \vee \sim r) \wedge \sim q]$

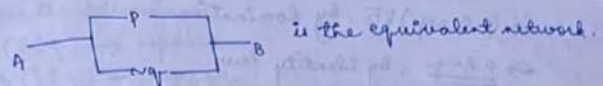
$\Leftrightarrow [p \wedge (q \vee \sim q) \vee \sim r)] \vee [(r \vee \sim r) \vee t] \wedge \sim q]$
by commutative & associative law.

$\Leftrightarrow [p \wedge (T_0 \vee \sim r)] \vee [(T_0 \vee \sim r) \wedge \sim q]$
inverse law.

$\Leftrightarrow [p \wedge (\sim r \vee T_0)] \vee [(\sim r \vee T_0) \wedge \sim q]$
commutative law.

$\Leftrightarrow [p \wedge T_0] \vee [T_0 \wedge \sim q]$, domination law.

$\Leftrightarrow p \vee \sim q$, identity law.



Simplify the following compound proposition using laws of logic.

$(p \vee q) \wedge [\sim(\sim p \wedge q)]$

Solⁿ - Given, $(p \vee q) \wedge [\sim(\sim p \wedge q)]$

$\Leftrightarrow (p \vee q) \wedge [\sim \sim p \vee \sim q]$, by De-Morgan's law.

$\Leftrightarrow (p \vee q) \wedge (p \vee \sim q)$, by double negation.

$\Leftrightarrow p \vee (q \wedge \sim q)$, by distributive law.

$\Leftrightarrow p \vee F_0$, by inverse law.

$\Leftrightarrow p$, by identity law.

Q) Simplify the following using laws of logic -
 $(p \vee q) \wedge [\sim(\sim p \vee q)]$

Solⁿ - Given, $(p \vee q) \wedge [\sim(\sim p \vee q)]$

$\Leftrightarrow (p \vee q) \wedge [\sim \sim p \wedge \sim q]$, by De-Morgan's law

$\Leftrightarrow (p \vee q) \wedge [p \wedge \sim q]$, double negation

$\Leftrightarrow [p \wedge (p \wedge \sim q) \wedge q \wedge (p \wedge \sim q)]$, by distributive law

$\Leftrightarrow [(p \wedge p) \wedge \sim q] \wedge [q \wedge \sim q] \wedge p$, by associative law

$\Leftrightarrow [C \wedge \sim q] \vee [F_0 \wedge p]$, by identity law.
by idempotent & inverse

$$\Leftrightarrow (p \wedge \sim q) \vee 0, \text{ by domination}$$

$$\Leftrightarrow \underline{p \wedge \sim q}, \text{ by identity law.}$$

Q) By using laws of logic, simplify -
HW $\sim[\sim(p \vee q) \wedge r] \vee \sim q$

* NAND and NOR Connectives:

→ The compound proposition $\sim(p \wedge q)$ is read as 'Not p and q' and represented by $p \uparrow q$.

→ The symbol $\uparrow$ is called NAND connective and it is the combination of NOT & AND.

→ The compound proposition $\sim(p \vee q)$ is read as 'Not p or q' and represented by $p \downarrow q$.

→ The symbol $\downarrow$ is called NOR connective and it is the combination of NOT & OR.

i.e, $p \uparrow q = \sim(p \wedge q) = \sim p \vee \sim q$

$p \downarrow q = \sim(p \vee q) = \sim p \wedge \sim q$

Truth table:

p	q	$p \wedge q$	$p \vee q$	$\sim(p \wedge q) = p \uparrow q$	$\sim(p \vee q) = p \downarrow q$
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	1	0
1	1	1	1	0	0

Q) Prove the following -

(i) $\sim(p \downarrow q) \Leftrightarrow \sim p \wedge \sim q$

(ii) $\sim(p \uparrow q) \Leftrightarrow \sim p \vee \sim q$

Solⁿ(i) - Consider LHS:

$$\sim(p \downarrow q) \Leftrightarrow \sim(\sim(p \vee q)), \text{ by definition of NOR.}$$

$$\Leftrightarrow \underline{\sim \sim p \vee \sim \sim q}, \text{ De Morgan's law.}$$

$$\Leftrightarrow \sim p \vee \sim q, \text{ by definition of NAND.}$$

Solⁿ(ii) Consider LHS:

$$\sim(p \uparrow q) \Leftrightarrow \sim(\sim(p \wedge q)), \text{ definition of NAND.}$$

$$\Leftrightarrow \sim(\sim p \vee \sim q), \text{ De Morgan's law.}$$

$$\Leftrightarrow \sim p \wedge \sim q, \text{ by definition of NOR.}$$

Q) For any 3 propositions p, q, r, prove the following -

(i) $p \uparrow (q \uparrow r) \Leftrightarrow \sim p \vee (q \wedge r)$

(ii) $(q \uparrow r) \uparrow p \Leftrightarrow (p \wedge q) \vee \sim r$

(iii) $p \downarrow (q \downarrow r) \Leftrightarrow \sim p \wedge (q \vee r)$

(iv) $(p \downarrow q) \downarrow r \Leftrightarrow (p \vee q) \wedge \sim r$

Solⁿ(i) LHS:

$$p \uparrow (q \uparrow r) \Leftrightarrow \sim(p \wedge (q \uparrow r)), \text{ by definition of NAND.}$$

$$\Leftrightarrow \sim(p \wedge \sim(q \wedge r)), \text{ by definition of NAND}$$

$$\Leftrightarrow \underline{\sim p \vee \sim \sim(q \wedge r)}, \text{ by De Morgan's law}$$

$$\Leftrightarrow \sim p \vee (q \wedge r), \text{ by double negation.}$$

Solⁿ(ii) LHS:

$$(p \uparrow q) \uparrow r$$

$$\Leftrightarrow \sim [\sim (p \wedge q) \wedge \sim r], \text{ definition of NAND}$$

$$\Leftrightarrow \sim (p \wedge q) \vee \sim r, \text{ De-Morgan's law}$$

$$\Leftrightarrow (p \wedge q) \vee r, \text{ Double negation}$$

Solⁿ(iii) LHS:

$$p \downarrow (q \downarrow r)$$

$$\Leftrightarrow \sim (p \vee \sim (q \vee r)), \text{ definition of NOR}$$

$$\Leftrightarrow \sim p \wedge \sim \sim (q \vee r), \text{ De-Morgan's law}$$

$$\Leftrightarrow \sim p \wedge (q \vee r), \text{ Double negation}$$

Solⁿ(iv) LHS:

$$(p \vee q) \wedge \sim (p \downarrow q) \downarrow r$$

$$\Leftrightarrow \sim [(p \vee q) \vee r], \text{ by definition of NOR}$$

$$\Leftrightarrow \sim \sim (p \vee q) \wedge \sim r, \text{ by De Morgan's law}$$

$$\Leftrightarrow (p \vee q) \wedge \sim r, \text{ by double negation}$$

HN

$$\text{Sol}^n = \sim [\sim \{ (p \vee q) \wedge r \} \vee \sim q]$$

$$\Leftrightarrow \sim [\sim (p \vee q) \vee \sim r] \vee \sim q, \text{ De Morgan's law}$$

$$\Leftrightarrow \sim [\sim p \wedge \sim q] \vee \sim r \vee \sim q, \text{ De-Morgan's law}$$

$$\Leftrightarrow \sim [\sim p \wedge \sim q] \vee \sim r \vee \sim q, \text{ Associative \& Commutative law}$$

$$\Leftrightarrow \sim [\sim q \vee \sim r], \text{ De-Morgan's Double negation}$$

$$\Leftrightarrow q \wedge r$$

* DUALITY:

Let 'S' be a compound proposition.

If S contains logical connectives $\wedge, \vee$ only & T_0, F_0 , then dual of S is denoted by S^d and it is the compound proposition obtained from S by replacing $\wedge$ by $\vee$, $\vee$ by $\wedge$, T_0 by F_0 & F_0 by T_0 respectively.

* PRINCIPLE OF DUALITY:

Let 'U' & 'V' be 2 compound propositions.

If $U \Leftrightarrow V$, then $U^d \Leftrightarrow V^d$.

This is called principle of duality.

Q) Write duals of the following compound propositions.

(i) $(p \vee q) \wedge r$

(ii) $(p \wedge q) \vee r$

(iii) $p \rightarrow q$

(iv) $q \rightarrow p$

(v) $(p \rightarrow q) \rightarrow r$

(vi) $p \rightarrow (q \rightarrow r)$

(vii) $p \rightarrow (q \wedge r)$

Soln-

i) $U = (p \wedge q) \vee r$

ii) $U^d = (p \vee q) \wedge \bar{r}$

iii) $U = p \rightarrow q$

$\Leftrightarrow \sim p \vee q$, using $p \rightarrow q \Leftrightarrow \sim p \vee q$

$U^d = \sim p \wedge q$

iv) $U = q \rightarrow p$

$\Leftrightarrow \sim q \vee p$, using $p \rightarrow q \Leftrightarrow \sim p \vee q$

$U^d = \sim q \wedge p$

v) $U = (p \rightarrow q) \rightarrow r$

$\Leftrightarrow \sim(p \rightarrow q) \vee r$, using $p \rightarrow q \Leftrightarrow \sim p \vee q$

$\Leftrightarrow \sim(\sim p \vee q) \vee r$

$\Leftrightarrow U^d = \sim(\sim p \wedge q) \wedge r$

vi) $U = p \rightarrow (q \rightarrow r)$

$\Leftrightarrow \sim p \vee (q \rightarrow r)$

$\Leftrightarrow \sim p \vee (\sim q \vee r)$, using $p \rightarrow q \Leftrightarrow \sim p \vee q$

$U^d = \sim p \wedge \sim q \wedge r$

vii) $U = p \rightarrow (q \wedge r)$

$\Leftrightarrow \sim p \vee (q \wedge r)$, using $p \rightarrow q \Leftrightarrow \sim p \vee q$

$U^d = \sim p \wedge \sim q \vee r$

q) verify the principle of duality for the following compound proposition using laws of logic.

$[(\sim p \wedge q) \rightarrow \sim p \vee (\sim p \vee q)] \Leftrightarrow \sim p \vee q$

Soln- Let us take,

$U = (\sim p \wedge q) \rightarrow \sim p \vee (\sim p \vee q)$

$V = \sim p \vee q$

$\therefore U^d = \sim p \wedge q \rightarrow (1)$

Consider $U = (\sim p \wedge q) \rightarrow \sim p \vee (\sim p \vee q)$

$\Leftrightarrow \sim[(\sim p \wedge q)] \vee \sim p \vee (\sim p \vee q)$, using $p \rightarrow q \Leftrightarrow \sim p \vee q$

$\Leftrightarrow p \wedge q \vee (\sim p \vee \sim p) \vee q$, by double negation, associative law.

$\Leftrightarrow (p \wedge q) \vee (\sim p \vee q)$, idempotent law.

$\Leftrightarrow [p \vee (\sim p \vee q)] \wedge [q \vee (\sim p \vee q)]$, distributive law.

$\Leftrightarrow [(p \vee \sim p) \vee q] \wedge [(q \vee \sim p) \vee q]$, associative & commutative law.

$\Leftrightarrow [T \wedge q] \vee p$, by inverse & idempotent law.

$\Leftrightarrow T \wedge (q \vee p)$, by domination law

$\Leftrightarrow q \vee p$, by identity law.

$U^d \Leftrightarrow \sim p \wedge q \rightarrow (2)$

$\therefore U^d = V^d$

From (3) & (1), we have,

$$U^d \leftrightarrow V^d$$

Hence, the principle of duality is verified

Q) Verify the duality for the following CP using laws of logic

$$[(\sim p \vee q) \wedge (p \wedge (p \wedge q))] \leftrightarrow p \wedge q$$

$$\text{Sol}^n - U \equiv [(\sim p \vee q) \wedge (p \wedge (p \wedge q))]$$

$$V \equiv p \wedge q$$

$$\therefore V^d = p \vee q \rightarrow (1)$$

$$U \equiv (\sim p \vee q) \wedge (p \wedge (p \wedge q))$$

$$\Leftrightarrow (\sim p \vee q) \wedge [(p \wedge p) \wedge q], \text{ By Associative law.}$$

$$\Leftrightarrow \sim p \vee q \wedge (p \wedge q), \text{ Idempotent law.}$$

$$\Leftrightarrow [\sim p \wedge (p \wedge q)] \vee [q \wedge (p \wedge q)], \text{ Distributive law.}$$

$$\Leftrightarrow [\sim p \wedge p] \wedge q \vee [q \wedge p] \wedge p, \text{ Commutative \& Associative}$$

$$\Leftrightarrow [0 \wedge q] \vee [q \wedge p], \text{ Inverse law.}$$

$$\Leftrightarrow 0 \vee (q \wedge p), \text{ Domination law.}$$

$$\Leftrightarrow q \wedge p, \text{ Identity law.}$$

$$\Leftrightarrow p \wedge q, \text{ Commutative law.}$$

$$U^d \leftrightarrow p \vee q \rightarrow (2)$$

From (2) & (1), we have,

$$U^d \leftrightarrow V^d$$

Hence, the principle of duality is verified

Q1) 01/25

* LOGICAL IMPLICATION:

2 propositions 'U' & 'V' are said to be logically implication if $U \rightarrow V$ is a tautology and we write $U \Rightarrow V$.

For eg. Prove that $[p \wedge (p \rightarrow q)] \Rightarrow q$ using truth table.

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	q
0	0	1	0	0
0	1	1	0	1
1	0	0	0	0
1	1	1	1	1

From the table, we note that - whenever $p \wedge (p \rightarrow q)$ is true, q is also true.

Q2) Prove using TT - $[(p \rightarrow q) \wedge \sim q] \Rightarrow \sim p$.

p	q	$p \rightarrow q$	$\sim q$	$(p \rightarrow q) \wedge \sim q$	$\sim p$
0	0	1	1	1	1
0	1	1	0	0	1
1	0	0	1	0	0
1	1	1	0	0	0

From the table, we note that - whenever $(p \rightarrow q) \wedge \sim q$ is true, $\sim p$ is also true.

$$93) (p \rightarrow q) \wedge (q \rightarrow r) \Rightarrow p \rightarrow r$$

	p	q	r	$p \rightarrow q$	$q \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r)$	$p \rightarrow r$
i	0	0	0	1	1	1	1
ii	0	0	1	1	1	1	1
	0	1	0	0	1	0	1
	0	1	1	1	1	1	1
	1	0	0	0	0	0	0
	1	0	1	0	1	0	0
	1	1	0	1	0	0	0
	1	1	1	1	1	1	1

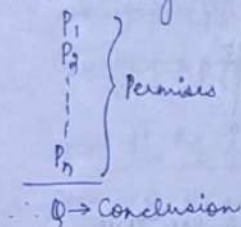
Hence, whenever P is true, Q is also true.

* RULES OF INFERENCE:

→ Consider a set of propositions $p_1, p_2, \dots, p_n$ and a proposition Q.
→ The compound proposition of the form $(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow Q$ is called an argument.

→ Here, $p_1, p_2, \dots, p_n$ are called statements or premises of the argument and Q is called conclusion of the argument.

→ We usually write the argument as follows-



→ In an argument, statements are always taken to be true, whereas, The conclusion may be true or false.

The conclusion is true only in the case of valid argument.
→ There exist rules of logic which can be used to check the validity of the argument.

→ These rules are called 'Rules of Inference'.

Rule-1: Rule of Conjunctive Simplification

$$p \wedge q \Rightarrow p \quad p \wedge q \Rightarrow q$$

Rule-2: Rule of Disjunctive Simplification

$$p \vee q \Rightarrow p \quad p \vee q \Rightarrow q$$

Rule-3: Rule of Syllogism

$$(p \rightarrow q) \wedge (q \rightarrow r) \Rightarrow (p \rightarrow r)$$

Rule-4: Modus Ponens rule

$$p \wedge (p \rightarrow q) \Rightarrow q$$

Rule-5: Modus Tollens rule

$$(p \rightarrow q) \wedge \neg q \Rightarrow \neg p$$

Rule-6: Rule of Disjunctive Syllogism

$$(p \vee q) \wedge \neg p \Rightarrow q$$

Q1) Test whether the following argument is valid or not -
If Virat hits a century, he gets a free car.

Virat hits a century. Therefore, Virat gets a free car.

Sol:- Let p: Virat hits a century
q: Virat gets a free car.

Argument: $p \rightarrow q$

$$\frac{p}{\therefore q}$$

23

CP: $(p \rightarrow q) \wedge p \Rightarrow r$

By using Modus Ponens rule, the given statement is valid.

Q2) If interest rate falls, then stock market will raise.
The stock market will not raise.
 $\therefore$ The interest rate will not fall.

Solⁿ- Let p: Interest rate falls
q: Stock market will raise

Argument: $p \rightarrow q$
 $\neg q$
 $\therefore \neg p$

CP: $(p \rightarrow q) \wedge \neg q \Rightarrow \neg p$

By using Modus Tollens rule, the given ^{argument} statement is valid.

Q3) Test whether the following argument is valid or not-

If I study then I don't fail in the examination.

If I don't fail in the examination then my father gifts a car to me.

$\therefore$ If I study then my father gifts a car to me

Solⁿ- Let p: I study
q: I don't fail in the exam
r: My Father gifts a car to me

Argument: $p \rightarrow q$
 $q \rightarrow r$
 $\therefore p \rightarrow r$

CP: $(p \rightarrow q) \wedge (q \rightarrow r) \Rightarrow p \rightarrow r$

By using rule of Syllogism, the argument is valid.

HW-

Q4) Test the validity of the argument-
If Sachin hits a century then he gets a free car.
Sachin does not get a free car.
 $\therefore$ Sachin does not hit a century.

Solⁿ- Let p: Sachin hits a century
q: Sachin gets a free car

Argument: $p \rightarrow q$
 $\neg q$
 $\therefore \neg p$

CP: $(p \rightarrow q) \wedge \neg q \Rightarrow \neg p$

By using ~~the~~ Modus Tollens rule, the argument is valid.

Q5) Test the validity of the following argument-
John will become famous or he will not become a musician.

John will become musician

$\therefore$ John will become famous.

Solⁿ- Let p: John will become famous
q: John will not become musician

Argument: $p \vee q$
 $\neg q$
 $\therefore p$

CP: $(p \vee q) \wedge \sim q \Rightarrow p$

by rule of disjunction syllogism

Consider LHS: $(p \vee q) \wedge \sim q$
 $\Leftrightarrow (\sim p \rightarrow q) \wedge \sim q$, using $p \rightarrow q \Leftrightarrow \sim p \vee q$
 $\Leftrightarrow \sim p$, by Modus Tollens
 $\Leftrightarrow p$, by double negation.
 $= \text{RHS}$

96) Test the validity -

If Ravi studies then he will pass DMGT.
 If Ravi does not play cricket then he will study.
 Ravi failed in DMGT
 $\therefore$ Ravi played cricket.

Solⁿ - Let p : Ravi studies

q : Ravi will pass DMGT

r : Ravi does not play cricket

Argument - $p \rightarrow q$

$$\begin{array}{l} r \rightarrow p \\ \hline \therefore \sim r \end{array}$$

CP: $(p \rightarrow q) \wedge (r \rightarrow p) \wedge \sim r \Rightarrow p$

Consider,

LHS: $(p \rightarrow q) \wedge (r \rightarrow p) \wedge \sim r$

$\Leftrightarrow (r \rightarrow p) \wedge (p \rightarrow q) \wedge \sim r$, By Commutative law

$\Leftrightarrow r \rightarrow q$, by Syllogism.

$\Leftrightarrow \sim r$, by Modus Tollens rule

$= \text{RHS}$

Hence, it is a valid argument.

97) Test the validity of the argument -

Ram will get DL or he will not get loan to buy a car.
 If Ram does not get loan then Ram will join as cab driver.

Ram got DL

$\therefore$ Ram will not join as cab driver.

Solⁿ - Let,

p : Ram will get DL

q : Ram will not get loan to buy a car

r : Ram will join as cab driver

Argument -

$$\begin{array}{l} p \vee q \\ q \rightarrow r \\ \hline p \\ \hline \sim r \end{array}$$

CP - $(p \vee q) \wedge (q \rightarrow r) \wedge p \Rightarrow \sim r$

Consider LHS: $(p \vee q) \wedge (q \rightarrow r) \wedge p$

$(p \vee q) \wedge (\sim q \rightarrow r) \wedge p$, using $p \rightarrow q \Leftrightarrow \sim p \vee q$

$\Leftrightarrow (\sim p \rightarrow r) \wedge (\sim q \rightarrow r) \wedge p$, using $p \rightarrow q \Leftrightarrow \sim p \vee q$ - (1)

$\Leftrightarrow (\sim p \rightarrow r) \wedge p$, ~~also~~ by Syllogism

$\Leftrightarrow [(\sim p) \vee r] \wedge p$, using (1)

$\Leftrightarrow (p \vee r) \wedge p$, by double negation

$\Leftrightarrow (p \wedge p) \vee (p \wedge r)$, distributive law

$\Leftrightarrow p \vee (p \wedge r)$, idempotent law

$\Leftrightarrow p \vee (p \wedge r)$, Commutative law

$\Leftrightarrow p$, by absorption law.

≠ RHS.

∴ It is not valid.

- (i) Check the validity of the following argument.
 If Anil gets the supervisor's position and works hard
 then he will get a raise.
 If he gets a raise then he will buy a new car.
 He has not bought a new car.
 ∴ Anil did not get the supervisor's position/
 he did not work hard.

Sol: Let p : Anil gets the supervisor's position.
 q : Anil works hard.
 r : Anil will get a raise.
 s : Anil will buy a new car.

Argument:
$$\begin{array}{l} (p \wedge q) \rightarrow r \\ r \rightarrow s \\ \hline \neg s \\ \hline \neg p \vee \neg q \end{array}$$

CP: $(p \wedge q) \rightarrow r \wedge (r \rightarrow s) \wedge \neg s \Rightarrow \neg p \vee \neg q$

Consider LHS:

$$\begin{aligned} & [(p \wedge q) \rightarrow r] \wedge (r \rightarrow s) \wedge \neg s \\ & \Leftrightarrow [(p \wedge q) \rightarrow r] \wedge \neg s \text{ by syllogism.} \\ & \Leftrightarrow \neg(p \wedge q), \text{ by Modus Tollens rule.} \\ & \Leftrightarrow \neg p \vee \neg q, \text{ by De-Morgan's Law} \\ & = \text{RHS.} \end{aligned}$$

∴ Argument is valid.

99) Test the validity -

(i)
$$\begin{array}{l} p \rightarrow q \\ r \rightarrow s \\ \hline p \vee q \wedge r \\ \hline q \vee s \end{array}$$

(ii)
$$\begin{array}{l} p \rightarrow q \\ r \rightarrow s \\ \hline \neg q \vee \neg s \\ \hline \neg(p \wedge r) \end{array}$$

Sol: CP - $(p \rightarrow q) \wedge (r \rightarrow s) \wedge (p \vee r)$
 $\Leftrightarrow (p \rightarrow q) \wedge (r \rightarrow s) \wedge (\neg p \vee r)$, Using $p \rightarrow q \Leftrightarrow \neg p \vee q$
 $\Leftrightarrow (p \rightarrow q) \wedge (\neg p \vee r) \wedge (r \rightarrow s)$, Using commutative law.
 $\Leftrightarrow (p \rightarrow q) \wedge (\neg p \rightarrow s)$, Syllogism
 $\Leftrightarrow (\neg q \rightarrow \neg p) \wedge (\neg p \rightarrow s)$, By Contradiction positive.
 $\Leftrightarrow (\neg q \rightarrow s)$, Syllogism
 $\Leftrightarrow \neg(\neg q) \wedge s$ Using ①.
 $\Leftrightarrow q \wedge s$, Double negation
 $= \text{RHS}$

∴ Argument is valid.

Sol: ii) CP - $(p \rightarrow q) \wedge (r \rightarrow s) \wedge (\neg q \vee \neg s) \Rightarrow \neg(p \wedge r)$

LHS:

$$\begin{aligned} & (p \rightarrow q) \wedge (r \rightarrow s) \wedge (\neg q \vee \neg s) \\ & \Leftrightarrow (p \rightarrow q) \wedge (r \rightarrow s) \wedge (\neg q \rightarrow \neg s), \text{ Using } p \rightarrow q \Leftrightarrow \neg p \vee q \text{ - ①} \\ & \Leftrightarrow (p \rightarrow q) \wedge (\neg s \rightarrow \neg r) \wedge (q \rightarrow \neg s), \text{ Using Contrapositive.} \\ & \Leftrightarrow (p \rightarrow q) \wedge (q \wedge \neg r), \text{ Syllogism} \\ & \Leftrightarrow (p \rightarrow \neg r), \text{ Syllogism.} \\ & \Leftrightarrow \neg p \wedge \neg r \text{ (Using ①)} \Leftrightarrow \neg(p \wedge r) \\ & = \text{RHS} \end{aligned}$$

∴ Argument is valid.

26/09/25

* OPEN SENTENCES / OPEN STATEMENTS:

- Consider the declarative statements x , $x+2=6$, $x=\sqrt{2}$, x is divisible by 10 etc, which are not propositions unless the symbol x is defined.
- Such statements are called open statements or open sentences.
- The undefined symbol present in the open statement such as x is called free variable.
- Open statements containing variable x are represented by $p(x)$, $q(x)$, $r(x)$,
- If U is a set of elements such that $p(x)$ becomes proposition when x is replaced by the elements of U .

Then, U is called 'Universe'.

If $a \in U$, then the proposition obtained by replacing x by a is denoted by $p(a)$.

★ POINTS TO REMEMBER:

- For any open statement $p(x)$, its negation is denoted by $\neg p(x)$.
- For open statements, $p(x)$ & $q(x)$ we have,
 - Conjunction $\rightarrow p(x) \wedge q(x)$
 - Disjunction $\rightarrow p(x) \vee q(x)$
 - Conditional $\rightarrow p(x) \rightarrow q(x)$
 - Bi-conditional $\rightarrow p(x) \leftrightarrow q(x)$
- For a conditional, $p(x) \rightarrow q(x)$, we have -
 - Converse: $q(x) \rightarrow p(x)$
 - Inverse: $\neg p(x) \rightarrow \neg q(x)$
 - Contrapositive: $\neg q(x) \rightarrow \neg p(x)$

Suppose the universal set consists of all integers. Consider the following open sentences -

$$p(x): x \leq 3$$

$$q(x): x+1 \text{ is odd}$$

$$r(x): x > 0$$

Write down the truth value of the following -

$$(i) p(-1) \wedge q(1)$$

$$(ii) \neg p(3) \vee r(1)$$

$$(iii) p(1) \leftrightarrow \neg q(2)$$

$$(iv) p(0) \rightarrow q(0)$$

Sol: (i)

$p(-1)$	$q(1)$	$p(-1) \wedge q(1)$
1	0	0

(ii)

$p(3)$	$\neg p(3)$	$r(1)$	$\neg p(3) \vee r(1)$
1	0	1	1

(iii)

$p(1)$	$q(2)$	$\neg q(2)$	$p(1) \leftrightarrow \neg q(2)$
1	1	0	0

(iv)

$p(0)$	$q(0)$	$p(0) \rightarrow q(0)$
1	1	1

* QUANTIFIERS:

The open sentences containing the words of the type - "all", "every", "some", "few", "each" and so on, which represents are called quantifiers.

There are 2 types of quantifiers, as follows -

Quantifier	Words	Notation
Universal Quantifier	for every/for any/ for all/for each	$\forall$
Existential Quantifier	Atleast one/ for some/there exists	$\exists$

- Quantified Statement -
 → A proposition involving the universal or existential quantifier is called quantified statement.
 → Thus, a quantified statement is a proposition of the form -
 $\forall x \in S, p(x),$
 $\exists x \in S, p(x)$

Rules of Inferences-

1) Rule of Universal specification - If $p(x)$ is an open statement and if $\forall x p(x)$ is true, then, $p(a)$ is true for each 'a' in the universe.

2) Rule of Universal generalization - If an open statement $p(x)$ is true for any arbitrary value 'a' from the universe, then, for all $\forall x p(x)$ is true.

3) Rule of Existential specification - If $\exists x p(x)$, is true then $p(a)$ is true for some 'a' in the universe.

4) Rule of Existential generalization - If $p(x)$ is true for some $x=a$ in the universe, then, $\exists x, p(x)$ is true.

5) Rule of negation - The negation of the quantified statement can be obtained by changing the quantifier from universal to existential and vice versa and also replace the open statement by its negation i.e.,

$$\neg[\forall x p(x)] \equiv \exists x, \neg p(x)$$

$$\neg[\exists x p(x)] \equiv \forall x, \neg p(x)$$

21/09/25

LOGICAL EQUIVALENCE:

→ 2 quantified statements are said to be logically equivalent whenever they have same truth values in all possible situations.

→ The following results holds good -

$$(i) \forall x [p(x) \wedge q(x)] \Leftrightarrow [\forall x p(x)] \wedge [\forall x q(x)]$$

$$(ii) \exists x [p(x) \vee q(x)] \Leftrightarrow [\exists x p(x)] \vee [\exists x q(x)]$$

$$(iii) \exists x [p(x) \rightarrow q(x)] \Leftrightarrow \exists x [\neg p(x) \vee q(x)]$$

$$(iv) \exists x \neg p(x) \Leftrightarrow \text{for some } x, \neg p(x)$$

LOGICAL IMPLICATION:

→ 2 quantified statements P & Q are said to be logically implied if Q is true whenever P is true and we write $P \Rightarrow Q$.

eg - For the universe of all integers

$$p(x) = x > 0$$

$$q(x) = x \text{ is even}$$

$$r(x) = x \text{ is a perfect square}$$

$$s(x) = x \text{ is divisible by 3}$$

$$t(x) = x \text{ is divisible by 7}$$

3) Write down the following statements in symbolic form -

(i) Atleast one integer is even.

$$\exists x q(x)$$

(ii) There exists a positive integer that is even.

$$\exists x [p(x) \wedge q(x)]$$

(iii) If x is even then x is not divisible by 3.
 $\forall x [q(x) \rightarrow \neg s(x)]$

when at least one thing exists it is not good, put $\forall$ (for all)

(iv) No even integer is divisible by 7.
 $\forall x [q(x) \rightarrow \neg t(x)]$

(v) There exists an even integer divisible by 3.
 $\exists x [q(x) \wedge s(x)]$

We can't use $\rightarrow$ because it will be true for every condition whereas in this $\exists x$ (only for some)

Q8) Express the statement of the following argument in symbolic form and check the validity.

All men are mortal

Sachin is a man

$\therefore$ Sachin is mortal

Sol- Let S be the set of all men

$p(x)$: x is a man

$q(x)$: x is mortal

a : Sachin

Argument is defined as-

$\forall x [p(x) \rightarrow q(x)]$

$p(a)$

$\therefore q(a)$

Compound proposition:

$\forall x [p(x) \rightarrow q(x)] \wedge p(a) \Rightarrow q(a)$

Consider-

LHS: $\forall x [p(x) \rightarrow q(x)] \wedge p(a)$

$\Rightarrow [p(a) \rightarrow q(a)] \wedge p(a)$, by rule of universal specification.

$\Rightarrow q(a)$, by Modus Ponens rule
 $= \text{RHS}$
 $\therefore$ It is a valid argument.

Q9) Using Quantifier method, find whether the following argument is valid.

(i) If a triangle has 2 equal sides then it is isosceles

(ii) If a triangle is isosceles then it has 2 equal angles

(iii) The triangle ABC does not have 2 equal angles

$\therefore$ ABC does not have 2 equal sides.

Sol- Let S be the set of all triangles

$p(x)$: x has 2 equal sides

$q(x)$: x is isosceles

$r(x)$: x has 2 equal angles

a : Triangle ABC

Argument is defined as

$\forall x [p(x) \rightarrow q(x)]$

$\forall x [q(x) \rightarrow r(x)]$

$\neg r(a)$

$\neg p(a)$

CP: $\forall x [p(x) \rightarrow q(x)] \wedge \forall x [q(x) \rightarrow r(x)] \wedge \neg r(a) \Rightarrow \neg p(a)$

Consider:

LHS:

$\forall x [p(x) \rightarrow q(x)] \wedge \forall x [q(x) \rightarrow r(x)] \wedge \neg r(a)$

$\Rightarrow [p(a) \rightarrow q(a)] \wedge [q(a) \rightarrow r(a)] \wedge \neg r(a)$, by rule of universal specification

$\Rightarrow [p(a) \rightarrow r(a)] \wedge \neg r(a)$, by Syllogism

$\Rightarrow \neg p(a)$, by Modus Tollens rule.

$= \text{RHS}$

$\therefore$ It is a valid argument.

Q) Find whether the following argument is valid or not.
 No Engineering students of 1st or 2nd semester studies logic.
 John is an Engineering student who studies logic.
 Therefore, John is not in 2nd semester.

Solⁿ - Let 'S' be the set of all Engineering students.

$p(x)$: x is in 1st semester

$q(x)$: x is in 2nd semester

$r(x)$: x studies logic

a : John

Argument is defined as -

$$\frac{\forall x [p(x) \vee q(x) \rightarrow \sim r(x)]}{r(a)}$$

$$\therefore \sim q(a)$$

CP: $\forall x [p(x) \vee q(x) \rightarrow \sim r(x)] \wedge r(a) \Rightarrow \sim q(a)$

Consider LHS -

$$[p(x) \vee q(x) \rightarrow \sim r(x)] \wedge r(a)$$

$$\Rightarrow [p(a) \vee q(a) \rightarrow \sim r(a)] \wedge r(a), \text{ By rule of Universal specification}$$

$$\Rightarrow r(a) \wedge [r(a) \rightarrow \sim(p(a) \vee q(a))], \text{ By Commutative \& Contrapositive}$$

$$\Rightarrow \sim(p(a) \vee q(a))$$

$$\Rightarrow \sim p(a) \wedge \sim q(a)$$

$$\Rightarrow \sim q(a).$$